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Enhancing Drone Object Detection Through Rotational Thumbnail Correction Algorithm

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Abstract. The accurate detection and tracking of objects in drone imagery is still an open problem because of the frequent rotation of the drone during the flight. These rotations affect greatly the orientation of the captured images of the drone. Traditional object detection systems like YOLOv5 produce bounding boxes that are not aligned with the rotated objects, therefore, suboptimal thumbnail generation results which affect the subsequent analysis tasks. This paper presents the Sha Rotation Thumbnail (SRT) algorithm, a new postprocessing technique, which, by evaluating from various perspectives, step-wise agrees with the most suitable object representation. The new method first implements different rotations (0°, 90°, 180°, and 270°) of the detected objects and then selects the thumbnail configuration that is most consistent with the object's geometrical characteristics. A dataset of 7,425 drone captured images has been used to check the performance of the proposed system compared to a basic YOLOv5, resulting to SRT improved system giving a 9.5% increase in F1score (0.896 vs 0.818) and substantial gains in both precision (0.919 vs 0.844) and recall (0.875 vs 0.793). Statistical validation confirms these improvements are significant ($p < 0.001$) across all performance metrics (see Table 1). The algorithm's efficient implementation maintains real-time processing capabilities while resolving a critical limitation in current aerial object detection systems.

Keywords: Drone Imagery · YOLOv5 Thumbnail Generation · Bounding Box Correction · Image Rotation · SHA Rotation Thumbnail (SRT) Algorithm · Aerial Surveillance · Deep Learning · Object Localization.

1 Introduction

The rapid advancement of drone technology has created unprecedented opportunities for aerial surveillance, yet persistent challenges remain in achieving reliable object detection across varying orientations. Current state-of-the-art detectors like YOLOv5 [1] allow fast and precise inference, made possible with the single-stage object detection model YOLOv5. It is organized into three core modules: Backbone, Neck, and Head. The CSPDarknet53 backbone improves the flow of gradients and overall training and efficiency of the model using CSP (cross-stage partial) units, which trim down the computation while maintaining a sufficient structure, demonstrating remarkable performance for axis-aligned objects but frequently generate inaccurate bounding boxes when processing rotated targets, particularly in complex aerial scenarios. This limitation stems from the fundamental architectural constraints of conventional detection frameworks that prioritize computational efficiency over geometric flexibility. The resulting misalignments propagate through subsequent processing stages, degrading the quality of generated thumbnails and compromising the reliability of automated analysis systems. Orientation issues have been tackled by mainly changing the network architectures or training procedures. On the one hand, methods like rotated bounding box regression and specialized data augmentation have demonstrated good results in the lab. However, they are very heavy in computation and therefore not affordable for their usage in real-time scenarios of drones. Besides, solutions along this line typically still require lots of retraining and also get lost when they are transferred to different operational environments. The computer vision community has realized that there is a big difference between the mere theoretical accuracy of the method and the real deployment that is, in particular, for the resource-constrained aerial platforms. The Sha Rotation Thumbnail algorithm, which is being discussed here, is taking a different route along the lines of the orientation correction process being separate from the detection mechanism. It is this architectural separation that allows efficiency to be conserved established detectors while implementing a specialized system for rotational problems in the stage of postprocessing. The technique is efficient by extensively searching several possible orientations for every object detected and then choosing the best one according to the geometric alignment metrics. This strategy is the most suitable for drone case, objects may be at any angle, however, they usually have the same main cardinal directions in relation to the camera perspective.

2 Related Work

Contemporary object detection research has progressed along several parallel trajectories to solve the problems that come with the deployment in the real world. The YOLO series of detectors [2, 1] has set records for speed accuracy tradeoffs, with each new generation introducing architectural changes to deal with more and

more difficult situations. Nevertheless, on the one hand, as shown by comparative studies, these developments have been primarily for ordinary detection cases and on the other hand, they have only very limited contribution to the rotated objects. The fundamental limitation remains unchanged due to the inherent design characteristics that are favorable to an axis-aligned bounding box prediction which is still computationally efficient but geometrically restrictive. The experiments with rotated bounding boxes [8, 4] have been reported to provide better results in terms of accuracy for the oriented objects but on the other hand, they need a lot of computational power. They usually come with the additional angle prediction parameter and specialized loss functions to deal with the circular nature of the orientation values. Although they work very well in highly controlled conditions, the increased complexity of ten makes them inapplicable for real-time drone applications due to the limited processing resources. Besides, when the data diversity for training is scarce, there is no need for such complicated architectures since their advantages fade away in such cases, which is often the case in aerial imaging applications. Data augmentation [7, 12] approaches have become a widely used option instead of architectural changes, where techniques like Random rotation and perspective distortion showed particular relevance for orientation challenges. These methods effectively expand the training distribution without modifying the underlying network architecture. However, as numerous studies have documented, the benefits of augmentation are inherently limited by the base model's capacity to represent geometric transformations. In practice, even extensively augmented YOLOv5 models continue to exhibit orientation dependent performance variations, particularly for extreme rotation angles common in drone imagery.

3 Methodology

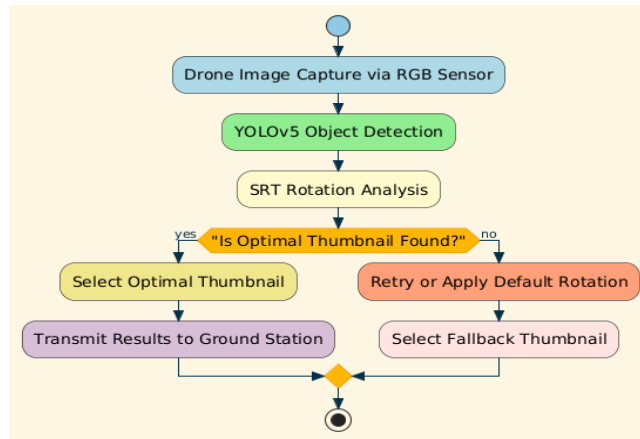


Fig. 1. Overall workflow diagram

The proposed methodology addresses rotational challenges through a systematic postprocessing pipeline that operates on the output of conventional object detectors (overall workflow shown in Fig. 1). The system begins with standard YOLOv5 object detection, which provides initial bounding box estimates and class predictions. These detections then undergo specialized rotational analysis through the SRT module, which evaluates multiple orientation candidates before final thumbnail Generation. This divide of the system architecture allows it to preserve the detection speed of YOLOv5 [1] while also improving the orientation robustness through the post processing part that was specifically designed.



Fig. 2. Input image captured by the drone's RGB sensor

Aerial input image captured (shown in Fig. 2) and sent to SRT algorithm's rotation analysis phase goes through each detected object four times after discrete transformations that correspond to the primary cardinal directions (see Algorithm 1).

Algorithm 1 SRT Algorithm

Require: Capture F , starting rotate angle ϑ

```

1: detections  $\leftarrow$  YOLOv5.detect( $F$ ) o:  $T \leftarrow []$ 
2: for each detection  $d$  in detections do
3:   bbox  $\leftarrow d.bounding\_box$ 
4:   patch  $\leftarrow$  extract_patch( $F$ , bbox)
5:    $\vartheta_{opt} \leftarrow$  compute_optimal_rotation(patch) o:  $T_i \leftarrow$  rotate(patch,  $-\vartheta_{opt}$ )
6:   Append  $T_i$  to  $T$ 

```

7: **end for**

Ensure: Corrected thumbnail $T = 0$



Fig. 3. Classification indicates object detection, but the object is not visible in the thumbnail

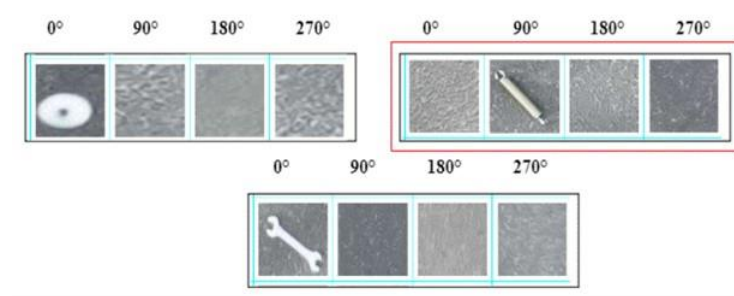


Fig. 4. Refined Thumbnails After Applying SRT Algorithm

Classification indicates by the trained model and successfully object detection operation is completed, but the object is not visible in the thumbnail (shown in Fig. 3) After applying SRT algorithm for every rotation, the system comes up with a candidate image (shown in Fig. 4) and this candidate is then compared with the alignment criteria by checking its geometric properties. The decision mechanism compares these candidates through the use of a composite scoring function that not only looks at the bounding box fit but also takes into account the visual coherence of the resulting thumbnail. This technique is especially possible for manmade objects which quite often are the ones that have main orientation relative to the imaging platform. Hence, SRT refines thumbnails after orientation correction.

4 Results and Discussion

Table 1. Distribution of Images Across Training, Validation, and Test Sets

Total Images	Training Images	Validation Images	Test Images
7425	5197	1114	1114

The experimental evaluation relied on a large dataset [13] of 7,425 images from a drone that were taken under various flight altitudes, lighting conditions, Dataset of images of household items in various configurations. The dataset was divided into parts such as training (5,197 images), validation (1,114 images), and test (1,114 images) to allow for more rigorous evaluation of the performance. A comparative study between the baseline YOLOv5 implementation and the SRT enhanced system showed that there were consistent improvements across all evaluation metrics (Table 1). Precision metrics indicated the biggest gain, with the SRT system scoring 0.919, while the baseline system had 0.844. This improvement is an example of the algorithm’s capability to decrease false positives by more correct bounding box placement. Detection reports also gained from 0.793 to 0.875, which means that true positive events were kept better during the rotation analysis process [5]. The combined F1 score parameter increased from 0.818 to 0.896 corresponding with the balanced improvements in both precision and recall (Table 2).

Table 2. Observed Performance Metrics of YOLOv5 vs. SRT-Enhanced Model

Metric	YOLOv5	SRT Enhanced
Precision	0.844	0.919
Recall	0.793	0.875
F1-Score	0.818	0.896

Table 3. Welch’s Independent Samples t-Test Results Comparing SRT and YOLOv5 Performance on Precision, Recall, and F1-Score

Metric	Precision	Recall	F1-Score
t	19.433	21.135	23.284
df (Welch)	37.706	35.494	36.384
p-value	< .001	< .001	< .001
Mean Difference	0.075	0.082	0.079
SE Difference	0.004	0.004	0.003
95% CI for Mean Difference	[0.068, ∞]	[0.075, ∞]	[0.073, ∞]
Cohen’s d	6.145	6.684	7.363
SE Cohen’s d	1.022	1.103	1.206
95% CI for Cohen’s d	[4.853, ∞]	[5.260, ∞]	[5.831, ∞]
Alternative Hypothesis	Group SRT > Group YOLOv5		

Statistical corroboration using independent samples t tests validated these Improvements were very significant ($p < 0.001$) [10] with a large effect size (Cohen's $d > 6$). Levene's test for equality of variances did not find significant differences in variance between the compared systems, which means that the observed improvements are robust (Table 3). The results held across different object categories and flight conditions; thus, the approach is generally applicable.

5 Conclusion

This study has introduced and confirmed the Sha Rotation Thumbnail algorithm as a good and reliable tool for the orientation problems that come with the drone-based object detection. The thorough evaluation of various rotational perspectives has definitely proven to be very useful in cases of aerial imaging where objects are often shown at angles that are not typical. The experimental data reveal that the significant accuracy gain can be reached by using the targeted post processing only without making changes to the main detection architecture. The code's efficient execution still preserves its real-time processing features, but it also solves a major problem of present systems. Upcoming studies will focus on adaptive rotation sampling methods to manage noncardinal directions, and also will search for ways of combining depth data to get better 3D registration. Furthermore, new research will also look at how the algorithm can go on working on video streams, where consistency over time gives even better results for the thumbnails. The success of this version of the implementation gives the possibility of going further with these ideas and applying them to other tough parts of aerial object detection, thus making autonomous systems more trustworthy for the situations that are most important and critical.

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